

PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Software Unit Test Report

Blood Bank Management System

Version 1.0 proved

Prepared by Team 9-Prakruti Prasanna Bhat

PES1UG24CS330

F-Section PES University

02/11/2026

Revision History

Version	Date	Details
1.0	02/11/2026	Initial report based on the completed BBMS test runs.

1. Introduction

1.1 Purpose

This report records the unit and component-level testing carried out for the Blood Bank Management System (BBMS). The tests check inventory calculations, validation of blood issue quantities, and role-based access rules.

1.2 Scope

Fourteen focused tests were run separately. The complete Django test suite was also executed to check that the wider project tests still passed. The full suite includes unit, integration, security, performance, and browser-related tests, so it is shown as supporting evidence rather than counted entirely as unit testing.

1.3 Reference Documents

BBMS Software Requirements Specification (SRS), Version 1.0.
BBMS Software Test Plan (STP), Version 1.0.
BBMS source code, Git revision 61d576d.

2. Test Environment

Parameter	Details
Operating system	Windows; Command Prompt
Python	3.14.6 (.venv)
Django	5.2.18
Testing tool	Django TestCase / test runner
Database	In-memory SQLite test database
Source revision	61d576d (GitHub main baseline)
System check	No issues reported (0 silenced)

Django created a temporary database for each run, applied the required migrations, and removed the test database after execution. The main application database was not used for these tests.

3. Testing Method

The selected tests use both valid and invalid inputs. Boundary values were included for blood quantity and expiry dates. Permission tests checked that the correct user role could access a protected function and that other roles were denied.

Some checks use Django models and a test database. For this reason, they are referred to as unit/component-level tests rather than fully isolated tests.

4. Focused Unit / Component Test Results

The following three test classes were executed together. All 14 tests passed.

Test class	Passed	Failed	Skipped
InventoryServiceTests	5	0	0
StockValidationTests	6	0	0
RoleRequiredDecoratorTests	3	0	0
Total	14	0	0

Execution time: 1.547 seconds. Result: OK. Process exit code: 0.

4.1 Individual Test Cases

ID	Check	Expected behavior	SRS reference	Status
UT-01	Available units exclude expired donations	Expired and pending units are not counted as available.	FR-4.5 / FR-5.3	PASS
UT-02	Inventory contains all blood groups	All eight standard blood groups appear.	FR-3.1 / DR-01	PASS
UT-03	Near-expiry units within seven days	A unit expiring in seven days is flagged; eight days is not.	FR-4.3	PASS
UT-04	Expired donations are reported	An expired unit appears in the expired-donation list.	FR-4.2	PASS
UT-05	Low stock below five units	Four units and zero units are flagged as low stock.	FR-4.4	PASS
UT-06	Issue allowed when stock is sufficient	Request for three units is allowed when five are available.	FR-7.3	PASS
UT-07	Issue allowed at available quantity	Request for five units is allowed when five are available.	FR-7.3	PASS
UT-08	Issue rejected for insufficient stock	Request for seven units is rejected when five are available.	FR-7.4	PASS
UT-09	Issue rejected when stock is zero	Request for unavailable A+ stock is rejected.	FR-7.4	PASS
UT-10	Zero requested units rejected	A zero-unit request raises a validation error.	Input validation	PASS
UT-11	Negative requested units rejected	A negative-unit request raises a validation error.	Input validation	PASS
UT-12	User profile created automatically	A profile is created for a new user.	Account setup	PASS
UT-13	Authorized staff role permitted	The protected view returns HTTP 200 for Staff.	FR-9.7 / SEC-02	PASS
UT-14	Hospital role denied staff access	The protected view denies Hospital Staff access.	FR-9.7 / SEC-02	PASS

5. Full Automated Suite Results

The full Django suite was run after the individual module tests. Django discovered 131 tests. Of these, 129 passed, and two were skipped. No test failed.

Module / test group	Tests found	Passed	Skipped
Authentication and accounts	6	6	0
Donor management	10	10	0
Donation recording	4	4	0

Module / test group	Tests found	Passed	Skipped
Inventory services	5	5	0
Blood availability search	10	10	0
Blood requests: submit/review	29	29	0
Blood requests: status tracking	12	12	0
Blood issue and fulfilment	40	39	1
Reports and history	13	13	0
Blood search performance	1	1	0
Browser portability	1	0	1
Total	131	129	2

Recorded result: Ran 131 tests in 68.791 seconds - OK (skipped=2). The process finished with exit code 0. The 14 focused checks are already included in these 131 tests.

6. Skipped Tests and Observations

Test	Reason recorded by Django
Concurrent blood issue test	Skipped because it requires PostgreSQL; the test run used SQLite.
Automated browser portability test	Skipped because Playwright browsers and the browser-test setting were not enabled.

Both tests were skipped, not passed. The browser checks carried out manually are separate from this automated run and should be recorded in the System Validation Report.

The test runs did not report any failures. The results do not indicate statement or branch coverage; those measurements belong in the separate coverage reports.

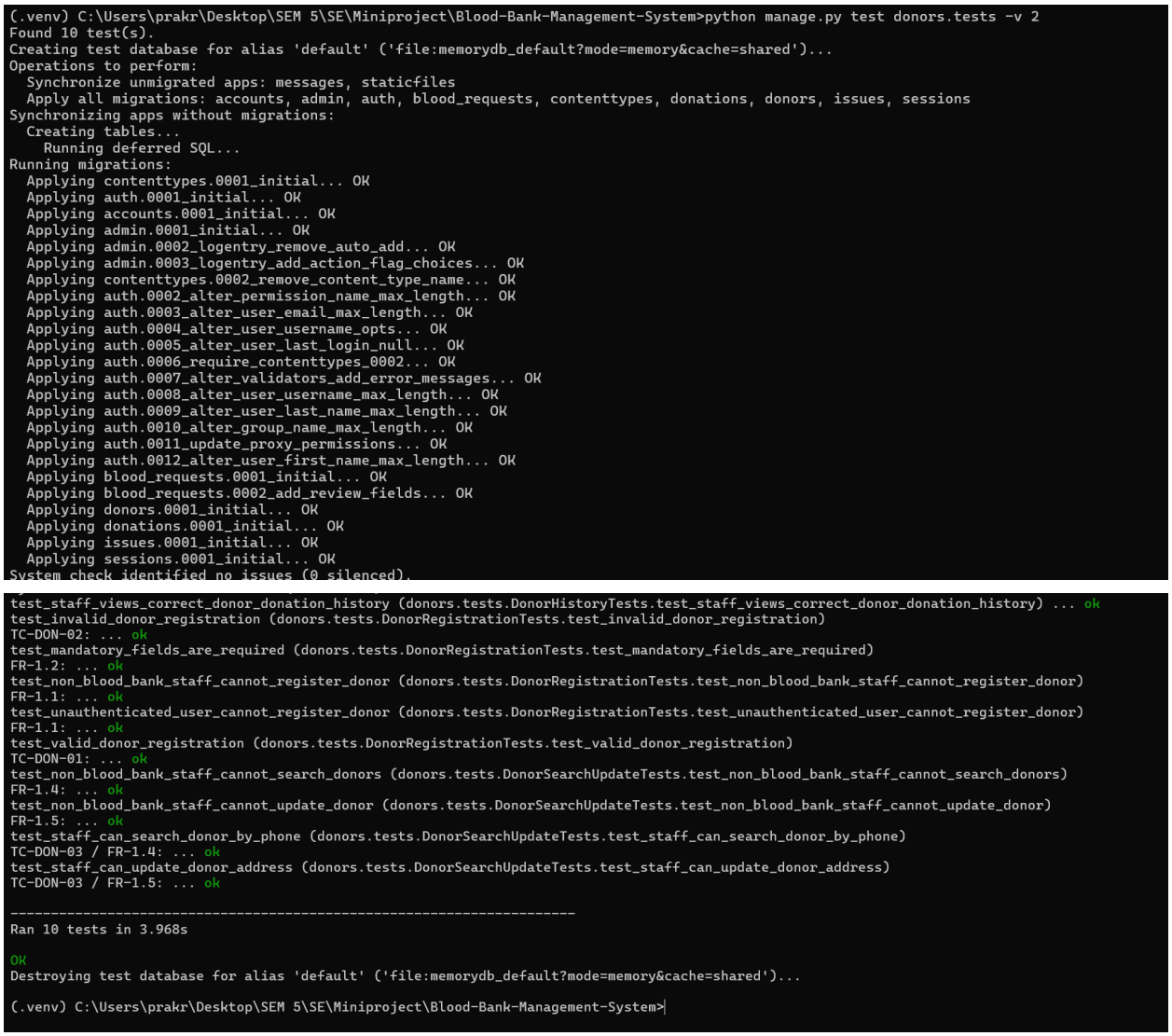
7. Conclusion

All 14 selected unit/component tests passed. The complete Django test suite also finished without failures: 129 tests passed and two were skipped for environment-related reasons. The results support the tested BBMS functions at revision 61d576d. PostgreSQL concurrency and automated browser testing were not verified in this run.

Appendix A. Test Execution Screenshots

The screenshots below are from the test commands run during verification on 5 November 2026. Each shows the relevant output and result summary. The complete command output was also saved as unit-test-execution.txt and unit-focused-execution.txt.

Figure A1. Donor management - 10 tests passed



```
(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>python manage.py test donors.tests -v 2
Found 10 test(s).
Creating test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles
  Apply all migrations: accounts, admin, auth, blood_requests, contenttypes, donations, donors, issues, sessions
Synchronizing apps without migrations:
  Creating tables...
    Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying accounts.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying blood_requests.0001_initial... OK
  Applying blood_requests.0002_add_review_fields... OK
  Applying donors.0001_initial... OK
  Applying donations.0001_initial... OK
  Applying issues.0001_initial... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
test_staff_views_correct_donor_donation_history (donors.tests.DonorHistoryTests.test_staff_views_correct_donor_donation_history) ... ok
test_invalid_donor_registration (donors.tests.DonorRegistrationTests.test_invalid_donor_registration)
TC-DON-02: ... ok
test_mandatory_fields_are_required (donors.tests.DonorRegistrationTests.test_mandatory_fields_are_required)
FR-1.2: ... ok
test_non_blood_bank_staff_cannot_register_donor (donors.tests.DonorRegistrationTests.test_non_blood_bank_staff_cannot_register_donor)
FR-1.1: ... ok
test_unauthenticated_user_cannot_register_donor (donors.tests.DonorRegistrationTests.test_unauthenticated_user_cannot_register_donor)
FR-1.1: ... ok
test_valid_donor_registration (donors.tests.DonorRegistrationTests.test_valid_donor_registration)
TC-DON-01: ... ok
test_non_blood_bank_staff_cannot_search_donors (donors.tests.DonorSearchUpdateTests.test_non_blood_bank_staff_cannot_search_donors)
FR-1.4: ... ok
test_non_blood_bank_staff_cannot_update_donor (donors.tests.DonorSearchUpdateTests.test_non_blood_bank_staff_cannot_update_donor)
FR-1.5: ... ok
test_staff_can_search_donor_by_phone (donors.tests.DonorSearchUpdateTests.test_staff_can_search_donor_by_phone)
TC-DON-03 / FR-1.4: ... ok
test_staff_can_update_donor_address (donors.tests.DonorSearchUpdateTests.test_staff_can_update_donor_address)
TC-DON-03 / FR-1.5: ... ok

----------------------------------------------------------------------
Ran 10 tests in 3.968s

OK
Destroying test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...

(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>
```

Figure A2. Donation recording - 4 tests passed

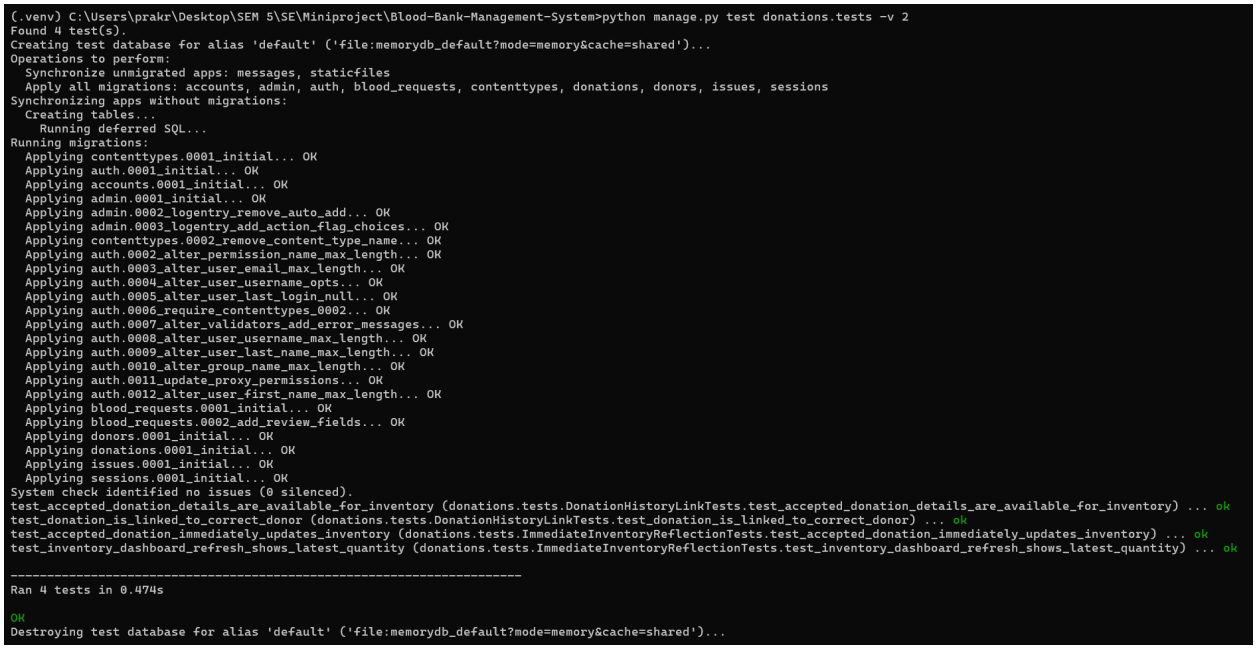


Figure A3. Blood inventory - 5 tests passed

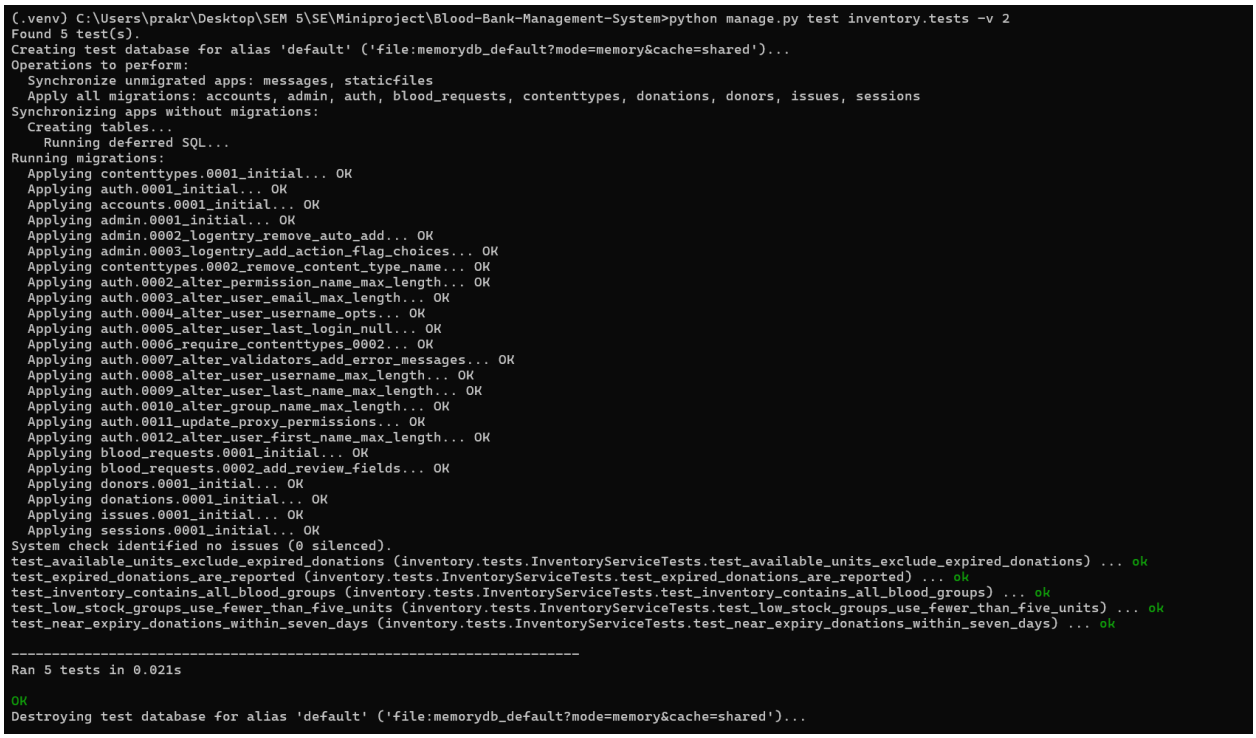
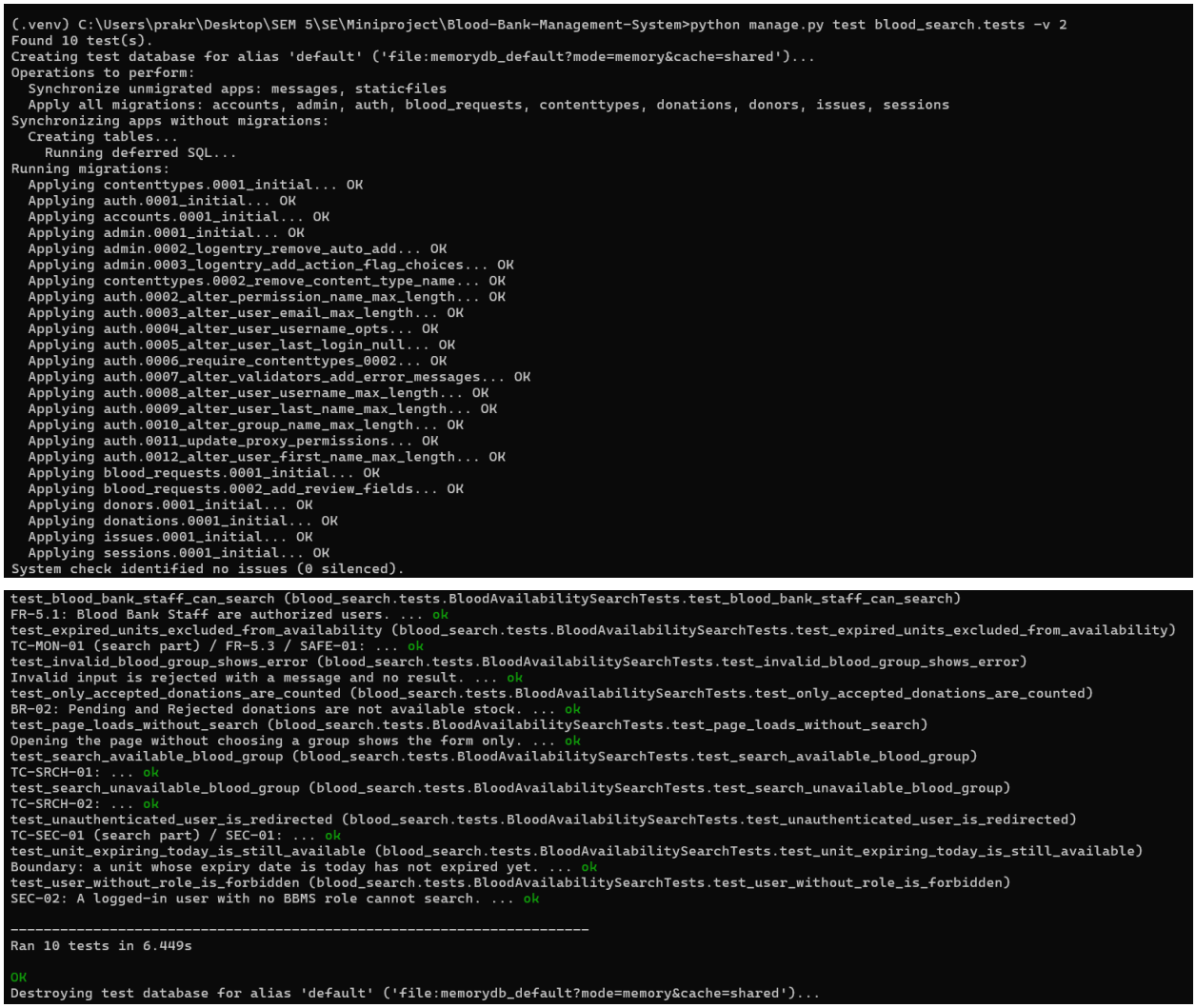


Figure A4. Blood availability search - 10 tests passed



```
(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>python manage.py test blood_search.tests -v 2
Found 10 test(s).
Creating test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles
  Apply all migrations: accounts, admin, auth, blood_requests, contenttypes, donations, donors, issues, sessions
Synchronizing apps without migrations:
  Creating tables...
    Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying accounts.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying blood_requests.0001_initial... OK
  Applying blood_requests.0002_add_review_fields... OK
  Applying donors.0001_initial... OK
  Applying donations.0001_initial... OK
  Applying issues.0001_initial... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
test_blood_bank_staff_can_search (blood_search.tests.BloodAvailabilitySearchTests.test_blood_bank_staff_can_search)
FR-5.1: Blood Bank Staff are authorized users. ... ok
test_expired_units_excluded_from_availability (blood_search.tests.BloodAvailabilitySearchTests.test_expired_units_excluded_from_availability)
TC-MON-01 (search part) / FR-5.3 / SAFE-01: ... ok
test_invalid_blood_group_shows_error (blood_search.tests.BloodAvailabilitySearchTests.test_invalid_blood_group_shows_error)
Invalid input is rejected with a message and no result. ... ok
test_only_accepted_donations_are_counted (blood_search.tests.BloodAvailabilitySearchTests.test_only_accepted_donations_are_counted)
BR-02: Pending and Rejected donations are not available stock. ... ok
test_page_loads_without_search (blood_search.tests.BloodAvailabilitySearchTests.test_page_loads_without_search)
Opening the page without choosing a group shows the form only. ... ok
test_search_available_blood_group (blood_search.tests.BloodAvailabilitySearchTests.test_search_available_blood_group)
TC-SRCH-01: ... ok
test_search_unavailable_blood_group (blood_search.tests.BloodAvailabilitySearchTests.test_search_unavailable_blood_group)
TC-SRCH-02: ... ok
test_unauthenticated_user_is_redirected (blood_search.tests.BloodAvailabilitySearchTests.test_unauthenticated_user_is_redirected)
TC-SEC-01 (search part) / SEC-01: ... ok
test_unit_expiring_today_is_still_available (blood_search.tests.BloodAvailabilitySearchTests.test_unit_expiring_today_is_still_available)
Boundary: a unit whose expiry date is today has not expired yet. ... ok
test_user_without_role_is_forbidden (blood_search.tests.BloodAvailabilitySearchTests.test_user_without_role_is_forbidden)
SEC-02: A logged-in user with no BBMS role cannot search. ... ok

----------------------------------------------------------------------
Ran 10 tests in 6.449s

OK
Destroying test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
```

Figure A5. Authentication - 6 tests passed

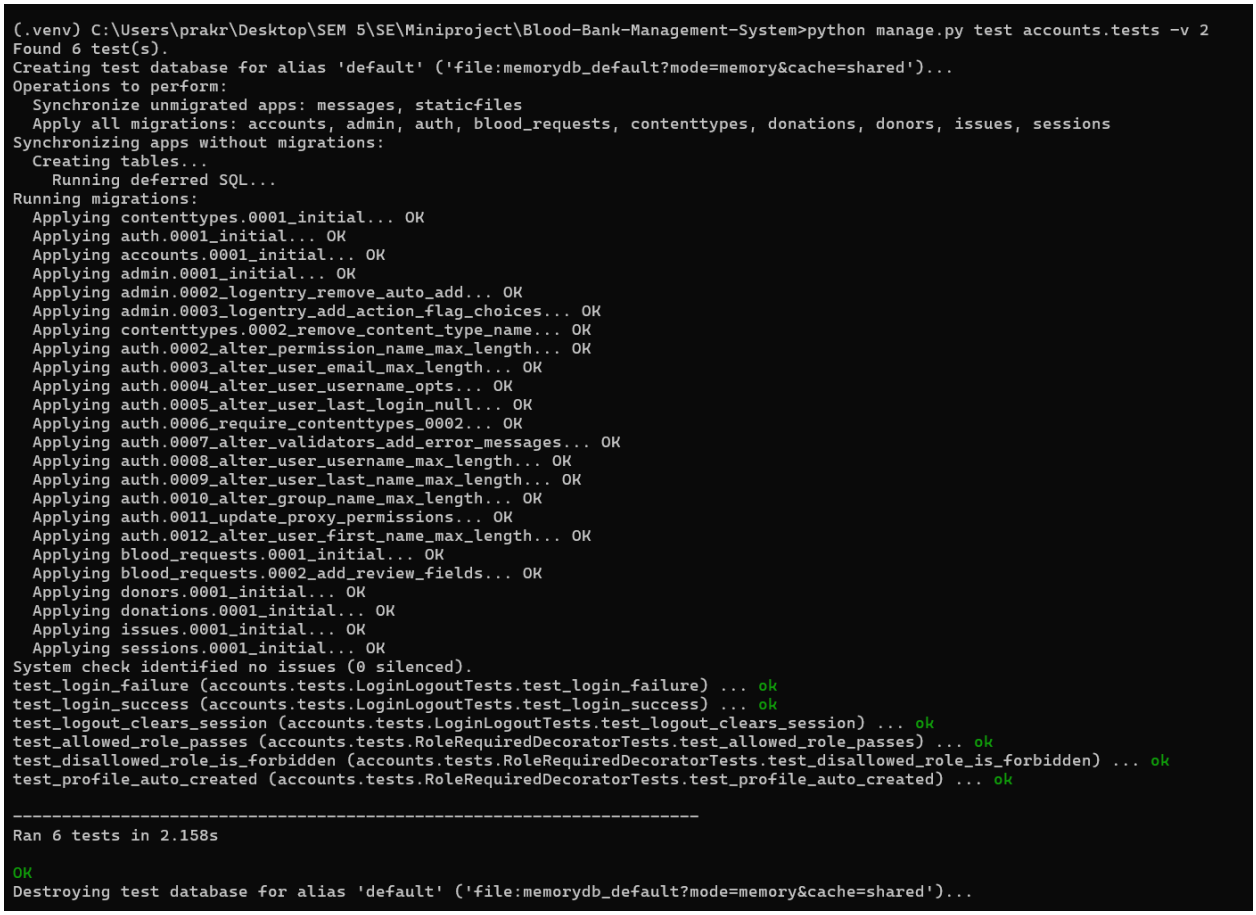


Figure A6. Blood requests - 29 tests passed



```
(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>python manage.py test blood_requests.tests -v 2
Found 29 test(s).
Creating test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles
  Apply all migrations: accounts, admin, auth, blood_requests, contenttypes, donations, donors, issues, sessions
Synchronizing apps without migrations:
  Creating tables...
    Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying accounts.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying blood_requests.0001_initial... OK
  Applying blood_requests.0002_add_review_fields... OK
  Applying donors.0001_initial... OK
  Applying donations.0001_initial... OK
  Applying issues.0001_initial... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
test_approved_request_leaves_pending_list (blood_requests.tests.ReviewBloodRequestTests.test_approved_request_leaves_pending_list) ... ok
test_hospital_user_cannot_approve_own_request (blood_requests.tests.ReviewBloodRequestTests.test_hospital_user_cannot_approve_own_request) ... ok
test_hospital_user_cannot_open_details (blood_requests.tests.ReviewBloodRequestTests.test_hospital_user_cannot_open_details) ... ok
test_hospital_user_cannot_open_pending_list (blood_requests.tests.ReviewBloodRequestTests.test_hospital_user_cannot_open_pending_list) ... ok
test_invalid_decision_changes_nothing (blood_requests.tests.ReviewBloodRequestTests.test_invalid_decision_changes_nothing) ... ok
test_missing_decision_changes_nothing (blood_requests.tests.ReviewBloodRequestTests.test_missing_decision_changes_nothing) ... ok
test_opening_details_does_not_change_status (blood_requests.tests.ReviewBloodRequestTests.test_opening_details_does_not_change_status) ... ok
test_request_cannot_be_reviewed_twice (blood_requests.tests.ReviewBloodRequestTests.test_request_cannot_be_reviewed_twice) ... ok
test_review_post_without_csrf_token_is_rejected (blood_requests.tests.ReviewBloodRequestTests.test_review_post_without_csrf_token_is_rejected) ... ok
test_staff_can_approve_request (blood_requests.tests.ReviewBloodRequestTests.test_staff_can_approve_request) ... ok
test_staff_can_open_request_details (blood_requests.tests.ReviewBloodRequestTests.test_staff_can_open_request_details) ... ok
test_staff_can_reject_request (blood_requests.tests.ReviewBloodRequestTests.test_staff_can_reject_request) ... ok
test_staff_sees_pending_request_in_list (blood_requests.tests.ReviewBloodRequestTests.test_staff_sees_pending_request_in_list) ... ok
test_unauthenticated_user_is_redirected_to_login (blood_requests.tests.ReviewBloodRequestTests.test_unauthenticated_user_is_redirected_to_login) ... ok
test_unknown_request_returns_404 (blood_requests.tests.ReviewBloodRequestTests.test_unknown_request_returns_404) ... ok
test_user_without_role_group_cannot_review (blood_requests.tests.ReviewBloodRequestTests.test_user_without_role_group_cannot_review)
A user whose automatically created profile defaults to STAFF ... ok
test_blood_bank_staff_cannot_submit_requests (blood_requests.tests.SubmitBloodRequestTests.test_blood_bank_staff_cannot_submit_requests)
FR-6.1 / SEC-02: only requesting organizations submit requests. ... ok
test_blood_group_is_prefilled_from_search (blood_requests.tests.SubmitBloodRequestTests.test_blood_group_is_prefilled_from_search)
Coming from Blood Search (?blood_group=O+) pre-selects the group. ... ok
test_each_request_gets_a_unique_id (blood_requests.tests.SubmitBloodRequestTests.test_each_request_gets_a_unique_id)
FR-6.3 / DR-04: each request is uniquely identified. ... ok
test_hospital_role_via_user_profile_can_submit (blood_requests.tests.SubmitBloodRequestTests.test_hospital_role_via_user_profile_can_submit)
Users set up with accounts.UserProfile role HOSPITAL ... ok
test_invalid_blood_group_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_invalid_blood_group_is_rejected)
DR-01: only standard blood groups are accepted. ... ok
test_missing_blood_group_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_missing_blood_group_is_rejected)
FR-6.7 / FR-6.2: blood group is mandatory. ... ok
test_missing_units_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_missing_units_is_rejected)
TC-REQ-02 / FR-6.7 / FR-6.2: ... ok
test_past_required_by_date_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_past_required_by_date_is_rejected)
A required-by date in the past is not accepted. ... ok
test_post_without_csrf_token_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_post_without_csrf_token_is_rejected)
TC-SEC-06 (request part) / SEC-07: ... ok
test_sql_injection_text_is_stored_as_plain_text (blood_requests.tests.SubmitBloodRequestTests.test_sql_injection_text_is_stored_as_plain_text)
TC-SEC-06 (request part) / SEC-08: ... ok
test_unauthenticated_user_is_redirected (blood_requests.tests.SubmitBloodRequestTests.test_unauthenticated_user_is_redirected)
TC-SEC-01 (request part) / SEC-01. ... ok
test_valid_request_is_created_as_pending (blood_requests.tests.SubmitBloodRequestTests.test_valid_request_is_created_as_pending)
TC-REQ-01 (submit part) / FR-6.1 – FR-6.4: ... ok
test_zero_units_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_zero_units_is_rejected)
Boundary: at least 1 unit must be requested. ... ok

----------------------------------------------------------------------
Ran 29 tests in 15.834s

OK
Destroying test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
```

Figure A7. Blood issuance - 39 passed, 1 skipped

```
(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>python manage.py test issues.tests -v 2
Found 40 test(s).
Creating test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles
  Apply all migrations: accounts, admin, auth, blood_requests, contenttypes, donations, donors, issues, sessions
Synchronizing apps without migrations:
  Creating tables...
    Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying accounts.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying blood_requests.0001_initial... OK
  Applying blood_requests.0002_add_review_fields... OK
  Applying donors.0001_initial... OK
  Applying donations.0001_initial... OK
  Applying issues.0001_initial... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
```

```
test_blood_search_stock_drops_after_issue (issues.tests.BloodSearchReflectsIssuedStockTests.test_blood_search_stock_drops_after_issue) ... ok
test_anonymous_user_gets_401_and_nothing_is_issued (issues.tests.IssueApiTests.test_anonymous_user_gets_401_and_nothing_is_issued) ... ok
test_get_is_not_allowed (issues.tests.IssueApiTests.test_get_is_not_allowed) ... ok
test_hospital_user_gets_403_and_nothing_is_issued (issues.tests.IssueApiTests.test_hospital_user_gets_403_and_nothing_is_issued) ... ok
test_insufficient_stock_returns_409_and_request_stays_approved (issues.tests.IssueApiTests.test_insufficient_stock_returns_409_and_request_stays_approved) ... ok
test_missing_request_id_returns_400 (issues.tests.IssueApiTests.test_missing_request_id_returns_400) ... ok
test_pending_request_returns_409_and_issues_nothing (issues.tests.IssueApiTests.test_pending_request_returns_409_and_issues_nothing) ... ok
test_route_is_exactly_api_issues (issues.tests.IssueApiTests.test_route_is_exactly_api_issues) ... ok
test_second_post_for_same_request_returns_409 (issues.tests.IssueApiTests.test_second_post_for_same_request_returns_409) ... ok
test_session_post_without_csrf_token_is_rejected (issues.tests.IssueApiTests.test_session_post_without_csrf_token_is_rejected) ... ok
test_staff_can_issue_with_form_data (issues.tests.IssueApiTests.test_staff_can_issue_with_form_data) ... ok
test_staff_can_issue_with_json_body (issues.tests.IssueApiTests.test_staff_can_issue_with_json_body) ... ok
test_unknown_request_returns_404 (issues.tests.IssueApiTests.test_unknown_request_returns_404) ... ok
test_allocation_failure_rolls_back_entire_transaction (issues.tests.IssueBloodServiceTests.test_allocation_failure_rolls_back_entire_transaction) ... ok
test_cannot_issue_the_same_request_twice (issues.tests.IssueBloodServiceTests.test_cannot_issue_the_same_request_twice) ... ok
test_expired_units_are_never_issued (issues.tests.IssueBloodServiceTests.test_expired_units_are_never_issued) ... ok
test_insufficient_stock_changes_nothing (issues.tests.IssueBloodServiceTests.test_insufficient_stock_changes_nothing) ... ok
test_issue_marks_request_fulfilled (issues.tests.IssueBloodServiceTests.test_issue_marks_request_fulfilled) ... ok
test_issue_records_who_issued_it (issues.tests.IssueBloodServiceTests.test_issue_records_who_issued_it) ... ok
test_issue_reduces_available_stock (issues.tests.IssueBloodServiceTests.test_issue_reduces_available_stock) ... ok
test_pending_donations_are_not_issued (issues.tests.IssueBloodServiceTests.test_pending_donations_are_not_issued) ... ok
test_pending_request_cannot_be_issued (issues.tests.IssueBloodServiceTests.test_pending_request_cannot_be_issued) ... ok
test_rejected_request_cannot_be_issued (issues.tests.IssueBloodServiceTests.test_rejected_request_cannot_be_issued) ... ok
test_stock_never_goes_negative_across_issues (issues.tests.IssueBloodServiceTests.test_stock_never_goes_negative_across_issues) ... ok
test_units_come_from_soonest_expiring_batch_first (issues.tests.IssueBloodServiceTests.test_units_come_from_soonest_expiring_batch_first) ... ok
test_anonymous_user_is_redirected_to_login (issues.tests.IssueViewTests.test_anonymous_user_is_redirected_to_login) ... ok
test_hospital_user_cannot_issue_blood (issues.tests.IssueViewTests.test_hospital_user_cannot_issue_blood) ... ok
test_hospital_user_cannot_open_issue_page (issues.tests.IssueViewTests.test_hospital_user_cannot_open_issue_page) ... ok
test_issue_page_shows_batches_after_issue (issues.tests.IssueViewTests.test_issue_page_shows_batches_after_issue) ... ok
test_opening_issue_page_does_not_issue_blood (issues.tests.IssueViewTests.test_opening_issue_page_does_not_issue_blood) ... ok
test_post_for_pending_request_shows_error_and_issues_nothing (issues.tests.IssueViewTests.test_post_for_pending_request_shows_error_and_issues_nothing) ... ok
test_staff_can_open_issue_page (issues.tests.IssueViewTests.test_staff_can_open_issue_page) ... ok
test_staff_post_issues_blood_and_redirects (issues.tests.IssueViewTests.test_staff_post_issues_blood_and_redirects) ... ok
test_allows_issue_when_requested_equals_available (issues.tests.StockValidationTests.test_allows_issue_when_requested_equals_available) ... ok
test_allows_issue_when_stock_is_sufficient (issues.tests.StockValidationTests.test_allows_issue_when_stock_is_sufficient) ... ok
test_rejects_issue_when_no_stock_is_available (issues.tests.StockValidationTests.test_rejects_issue_when_no_stock_is_available) ... ok
test_rejects_issue_when_stock_is_insufficient (issues.tests.StockValidationTests.test_rejects_issue_when_stock_is_insufficient) ... ok
test_rejects_negative_requested_units (issues.tests.StockValidationTests.test_rejects_negative_requested_units) ... ok
test_rejects_zero_requested_units (issues.tests.StockValidationTests.test_rejects_zero_requested_units) ... ok
test_concurrent_issues_do_not_overspend_stock (issues.tests.ConcurrentIssuanceTests.test_concurrent_issues_do_not_overspend_stock) ... skipped 'This concurrency test requ
ires PostgreSQL.'

----------------------------------------------------------------------
Ran 40 tests in 24.643s

OK (skipped=1)
Destroying test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
```

Figure A8. Reports and history - 13 tests passed

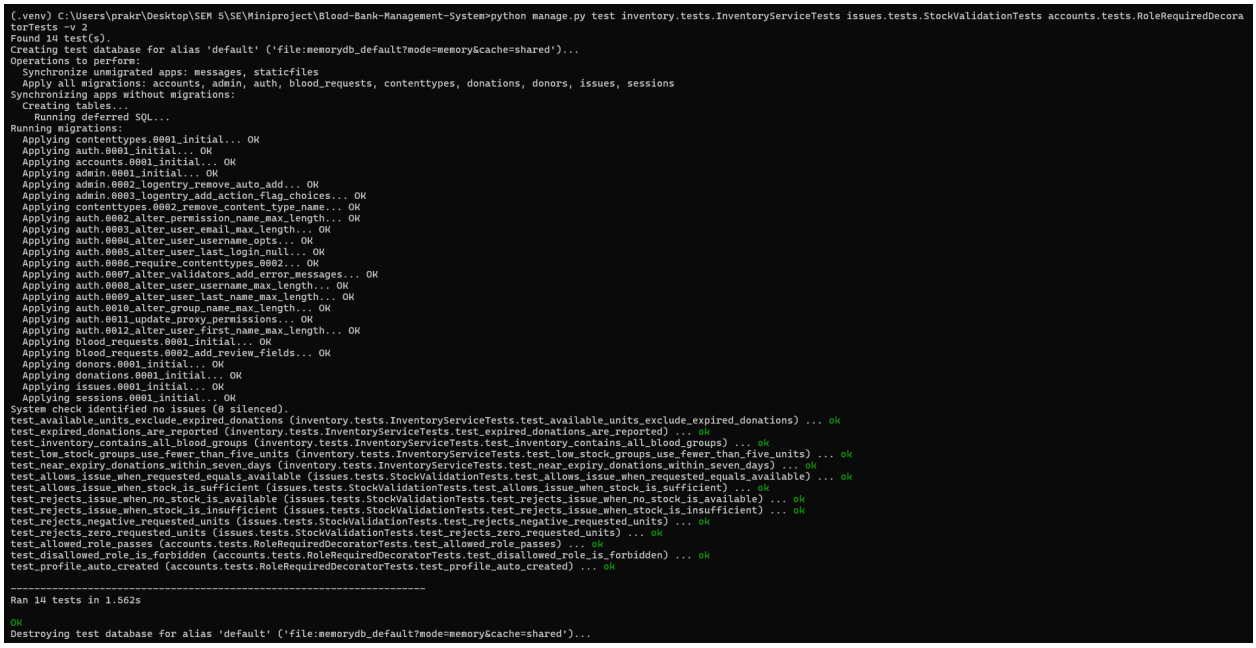


```
(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>python manage.py test reports.tests -v 2
Found 13 test(s).
Creating test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles
  Apply all migrations: accounts, admin, auth, blood_requests, contenttypes, donations, donors, issues, sessions
Synchronizing apps without migrations:
  Creating tables...
    Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying accounts.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying blood_requests.0001_initial... OK
  Applying blood_requests.0002_add_review_fields... OK
  Applying donors.0001_initial... OK
  Applying donations.0001_initial... OK
  Applying issues.0001_initial... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
test_administrator_can_open_reports (reports.tests.ReportsAccessTests.test_administrator_can_open_reports) ... ok
test_anonymous_user_is_redirected_to_login (reports.tests.ReportsAccessTests.test_anonymous_user_is_redirected_to_login) ... ok
test_blood_bank_staff_can_open_reports (reports.tests.ReportsAccessTests.test_blood_bank_staff_can_open_reports) ... ok
test_hospital_user_cannot_open_reports (reports.tests.ReportsAccessTests.test_hospital_user_cannot_open_reports) ... ok
test_group_row_shows_current_stock_after_issue (reports.tests.ReportsContentTests.test_group_row_shows_current_stock_after_issue) ... ok
test_issue_history_lists_the_issue (reports.tests.ReportsContentTests.test_issue_history_lists_the_issue) ... ok
test_patient_names_are_not_shown (reports.tests.ReportsContentTests.test_patient_names_are_not_shown) ... ok
test_request_history_shows_status (reports.tests.ReportsContentTests.test_request_history_shows_status) ... ok
test_summary_counts_donated_and_issued_units (reports.tests.ReportsContentTests.test_summary_counts_donated_and_issued_units) ... ok
test_blood_group_filter (reports.tests.ReportsFilterTests.test_blood_group_filter) ... ok
test_donations_outside_date_range_are_excluded (reports.tests.ReportsFilterTests.test_donations_outside_date_range_are_excluded) ... ok
test_end_date_before_start_date_shows_error (reports.tests.ReportsFilterTests.test_end_date_before_start_date_shows_error) ... ok
test_request_status_filter (reports.tests.ReportsFilterTests.test_request_status_filter) ... ok

----------------------------------------------------------------------
Ran 13 tests in 9.448s

OK
Destroying test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
```

Figure A9. Focused run - 14 tests passed

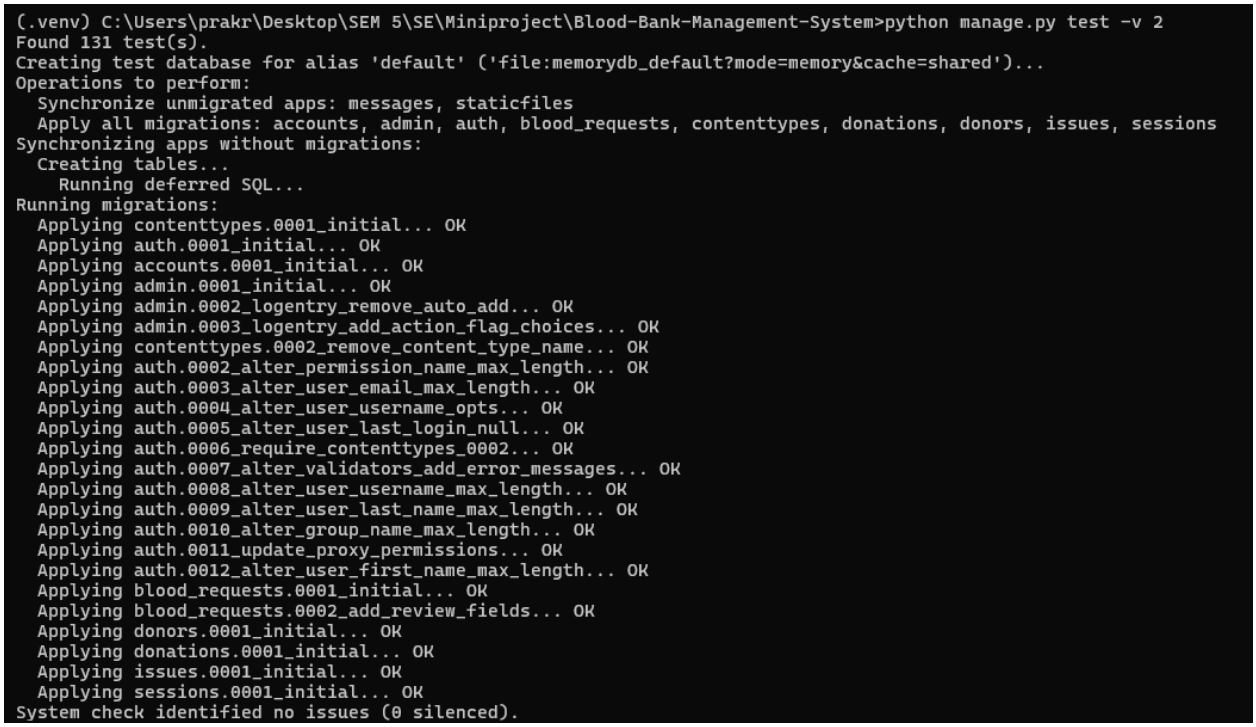


```
(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>python manage.py test inventory.tests.InventoryServiceTests issues.tests.StockValidationTests accounts.tests.RoleRequiredDecora
torTests -v 2
Found 14 test(s).
Creating test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles
  Apply all migrations: accounts, admin, auth, blood_requests, contenttypes, donations, donors, issues, sessions
Synchronizing apps without migrations:
  Creating tables...
    Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying accounts.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying blood_requests.0001_initial... OK
  Applying blood_requests.0002_add_review_fields... OK
  Applying donors.0001_initial... OK
  Applying donations.0001_initial... OK
  Applying issues.0001_initial... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
test_available_units_exclude_expired_donations (inventory.tests.InventoryServiceTests.test_available_units_exclude_expired_donations) ... ok
test_expired_donations_are_reported (inventory.tests.InventoryServiceTests.test_expired_donations_are_reported) ... ok
test_inventory_contains_all_blood_groups (inventory.tests.InventoryServiceTests.test_inventory_contains_all_blood_groups) ... ok
test_low_stock_groups_use_fewer_than_five_units (inventory.tests.InventoryServiceTests.test_low_stock_groups_use_fewer_than_five_units) ... ok
test_near_expiry_donations_within_seven_days (inventory.tests.InventoryServiceTests.test_near_expiry_donations_within_seven_days) ... ok
test_allows_issue_when_requested_equals_available (issues.tests.StockValidationTests.test_allows_issue_when_requested_equals_available) ... ok
test_allows_issue_when_stock_is_sufficient (issues.tests.StockValidationTests.test_allows_issue_when_stock_is_sufficient) ... ok
test_rejects_issue_when_no_stock_is_available (issues.tests.StockValidationTests.test_rejects_issue_when_no_stock_is_available) ... ok
test_rejects_issue_when_stock_is_insufficient (issues.tests.StockValidationTests.test_rejects_issue_when_stock_is_insufficient) ... ok
test_rejects_negative_requested_units (issues.tests.StockValidationTests.test_rejects_negative_requested_units) ... ok
test_rejects_zero_requested_units (issues.tests.StockValidationTests.test_rejects_zero_requested_units) ... ok
test_allowed_role_passes (accounts.tests.RoleRequiredDecoratorTests.test_allowed_role_passes) ... ok
test_disallowed_role_is_forbidden (accounts.tests.RoleRequiredDecoratorTests.test_disallowed_role_is_forbidden) ... ok
test_profile_auto_created (accounts.tests.RoleRequiredDecoratorTests.test_profile_auto_created) ... ok

----------------------------------------------------------------------
Ran 14 tests in 1.562s

OK
Destroying test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
```

Figure A10. Full-suite run - 129 passed, 2 skipped



```
(.venv) C:\Users\prakr\Desktop\SEM 5\SE\Miniproject\Blood-Bank-Management-System>python manage.py test -v 2
Found 131 test(s).
Creating test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles
  Apply all migrations: accounts, admin, auth, blood_requests, contenttypes, donations, donors, issues, sessions
Synchronizing apps without migrations:
  Creating tables...
    Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying accounts.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying blood_requests.0001_initial... OK
  Applying blood_requests.0002_add_review_fields... OK
  Applying donors.0001_initial... OK
  Applying donations.0001_initial... OK
  Applying issues.0001_initial... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
```



```
test_login_failure (accounts.tests.LoginLogoutTests.test_login_failure) ... ok
test_login_success (accounts.tests.LoginLogoutTests.test_login_success) ... ok
test_logout_clears_session (accounts.tests.LoginLogoutTests.test_logout_clears_session) ... ok
test_allowed_role_passes (accounts.tests.RoleRequiredDecoratorTests.test_allowed_role_passes) ... ok
test_disallowed_role_is_forbidden (accounts.tests.RoleRequiredDecoratorTests.test_disallowed_role_is_forbidden) ... ok
test_profile_auto_created (accounts.tests.RoleRequiredDecoratorTests.test_profile_auto_created) ... ok
test_hospital_cannot_open_all_requests (blood_requests.test_tracking.TrackRequestStatusTests.test_hospital_cannot_open_all_requests)
TC-SEC-02 / SEC-02: Hospital Staff cannot see other hospitals' ... ok
test_hospital_does_not_see_other_hospitals_requests (blood_requests.test_tracking.TrackRequestStatusTests.test_hospital_does_not_see_other_hospitals_requests)
SEC-03 / SO-01: users only see their own requests. ... ok
test_hospital_sees_own_requests_with_status (blood_requests.test_tracking.TrackRequestStatusTests.test_hospital_sees_own_requests_with_status)
TC-REQ-01 (track part) / FR-6.6: ... ok
test_my_requests_status_filter (blood_requests.test_tracking.TrackRequestStatusTests.test_my_requests_status_filter)
Filtering My Requests by status shows only matching requests. ... ok
test_script_in_review_note_is_escaped (blood_requests.test_tracking.TrackRequestStatusTests.test_script_in_review_note_is_escaped)
TC-SEC-06 (track part) / SEC-09: text is shown, never executed. ... ok
test_staff_all_requests_status_filter (blood_requests.test_tracking.TrackRequestStatusTests.test_staff_all_requests_status_filter) ... ok
test_staff_cannot_open_my_requests (blood_requests.test_tracking.TrackRequestStatusTests.test_staff_cannot_open_my_requests) ... ok
test_staff_sees_all_requests_from_all_hospitals (blood_requests.test_tracking.TrackRequestStatusTests.test_staff_sees_all_requests_from_all_hospitals)
TC-REQ-01 step 5 / FR-6.5: ... ok
test_status_reflects_approval_and_rejection (blood_requests.test_tracking.TrackRequestStatusTests.test_status_reflects_approval_and_rejection)
TC-ISS-01 (status part) / FR-6.6: ... ok
test_submit_redirects_to_my_requests_showing_pending (blood_requests.test_tracking.TrackRequestStatusTests.test_submit_redirects_to_my_requests_showing_pending)
TC-REQ-01 steps 3-4: after submitting, the user lands on ... ok
test_unauthenticated_users_are_redirected (blood_requests.test_tracking.TrackRequestStatusTests.test_unauthenticated_users_are_redirected)
TC-SEC-01 / SEC-01. ... ok
test_unknown_status_filter_is_ignored (blood_requests.test_tracking.TrackRequestStatusTests.test_unknown_status_filter_is_ignored) ... ok
test_approved_request_leaves_pending_list (blood_requests.tests.ReviewBloodRequestTests.test_approved_request_leaves_pending_list) ... ok
test_hospital_user_cannot_approve_own_request (blood_requests.tests.ReviewBloodRequestTests.test_hospital_user_cannot_approve_own_request) ... ok
test_hospital_user_cannot_open_details (blood_requests.tests.ReviewBloodRequestTests.test_hospital_user_cannot_open_details) ... ok
test_hospital_user_cannot_open_pending_list (blood_requests.tests.ReviewBloodRequestTests.test_hospital_user_cannot_open_pending_list) ... ok
test_invalid_decision_changes_nothing (blood_requests.tests.ReviewBloodRequestTests.test_invalid_decision_changes_nothing) ... ok
test_missing_decision_changes_nothing (blood_requests.tests.ReviewBloodRequestTests.test_missing_decision_changes_nothing) ... ok
test_opening_details_does_not_change_status (blood_requests.tests.ReviewBloodRequestTests.test_opening_details_does_not_change_status) ... ok
test_request_cannot_be_reviewed_twice (blood_requests.tests.ReviewBloodRequestTests.test_request_cannot_be_reviewed_twice) ... ok
test_review_post_without_csrf_token_is_rejected (blood_requests.tests.ReviewBloodRequestTests.test_review_post_without_csrf_token_is_rejected) ... ok
test_staff_can_approve_request (blood_requests.tests.ReviewBloodRequestTests.test_staff_can_approve_request) ... ok
test_staff_can_open_request_details (blood_requests.tests.ReviewBloodRequestTests.test_staff_can_open_request_details) ... ok
test_staff_can_reject_request (blood_requests.tests.ReviewBloodRequestTests.test_staff_can_reject_request) ... ok
test_staff_sees_pending_request_in_list (blood_requests.tests.ReviewBloodRequestTests.test_staff_sees_pending_request_in_list) ... ok
test_unauthenticated_user_is_redirected_to_login (blood_requests.tests.ReviewBloodRequestTests.test_unauthenticated_user_is_redirected_to_login) ... ok
test_unknown_request_returns_404 (blood_requests.tests.ReviewBloodRequestTests.test_unknown_request_returns_404) ... ok
test_user_without_role_group_cannot_review (blood_requests.tests.ReviewBloodRequestTests.test_user_without_role_group_cannot_review)
A user whose automatically created profile defaults to STAFF ... ok
test_blood_bank_staff_cannot_submit_requests (blood_requests.tests.SubmitBloodRequestTests.test_blood_bank_staff_cannot_submit_requests)
FR-6.1 / SEC-02: only requesting organizations submit requests. ... ok
test_blood_group_is_prefilled_from_search (blood_requests.tests.SubmitBloodRequestTests.test_blood_group_is_prefilled_from_search)
Coming from Blood Search (?blood_group=O+) pre-selects the group. ... ok
test_each_request_gets_a_unique_id (blood_requests.tests.SubmitBloodRequestTests.test_each_request_gets_a_unique_id)
FR-6.3 / DR-04: each request is uniquely identified. ... ok
test_hospital_role_via_user_profile_can_submit (blood_requests.tests.SubmitBloodRequestTests.test_hospital_role_via_user_profile_can_submit)
Users set up with accounts.UserProfile role HOSPITAL ... ok
test_invalid_blood_group_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_invalid_blood_group_is_rejected)
DR-01: only standard blood groups are accepted. ... ok
test_missing_blood_group_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_missing_blood_group_is_rejected)
FR-6.7 / FR-6.2: blood group is mandatory. ... ok
test_missing_units_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_missing_units_is_rejected)
TC-REQ-02 / FR-6.7 / FR-6.2: ... ok
test_past_required_by_date_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_past_required_by_date_is_rejected)
A required-by date in the past is not accepted. ... ok
test_post_without_csrf_token_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_post_without_csrf_token_is_rejected)
TC-SEC-06 (request part) / SEC-07: ... ok
```

```
test_sql_injection_text_is_stored_as_plain_text (blood_requests.tests.SubmitBloodRequestTests.test_sql_injection_text_is_stored_as_plain_text)
TC-SEC-06 (request part) / SEC-08: ... ok
test_unauthenticated_user_is_redirected (blood_requests.tests.SubmitBloodRequestTests.test_unauthenticated_user_is_redirected)
TC-SEC-01 (request part) / SEC-01. ... ok
test_valid_request_is_created_as_pending (blood_requests.tests.SubmitBloodRequestTests.test_valid_request_is_created_as_pending)
TC-REQ-01 (submit part) / FR-6.1 – FR-6.4: ... ok
test_zero_units_is_rejected (blood_requests.tests.SubmitBloodRequestTests.test_zero_units_is_rejected)
Boundary: at least 1 unit must be requested. ... ok
test_availability_search_responds_within_3_seconds (blood_search.test_performance.BloodAvailabilitySearchPerformanceTests.test_availability_search_responds_within_3_seconds)
TC-PERF-01 / NFR-01: ...
[TC-PERF-01] Availability search response times (s): A+=0.006, A−=0.005, B+=0.005, B−=0.005, AB+=0.005, AB−=0.005, O+=0.005, O−=0.005 | slowest: A+ 0.006s
ok
test_blood_bank_staff_can_search (blood_search.tests.BloodAvailabilitySearchTests.test_blood_bank_staff_can_search)
FR-5.1: Blood Bank Staff are authorized users. ... ok
test_expired_units_excluded_from_availability (blood_search.tests.BloodAvailabilitySearchTests.test_expired_units_excluded_from_availability)
TC-MON-01 (search part) / FR-5.3 / SAFE-01: ... ok
test_invalid_blood_group_shows_error (blood_search.tests.BloodAvailabilitySearchTests.test_invalid_blood_group_shows_error)
Invalid input is rejected with a message and no result. ... ok
test_only_accepted_donations_are_counted (blood_search.tests.BloodAvailabilitySearchTests.test_only_accepted_donations_are_counted)
BR-02: Pending and Rejected donations are not available stock. ... ok
test_page_loads_without_search (blood_search.tests.BloodAvailabilitySearchTests.test_page_loads_without_search)
Opening the page without choosing a group shows the form only. ... ok
test_search_available_blood_group (blood_search.tests.BloodAvailabilitySearchTests.test_search_available_blood_group)
TC-SRCH-01: ... ok
test_search_unavailable_blood_group (blood_search.tests.BloodAvailabilitySearchTests.test_search_unavailable_blood_group)
TC-SRCH-02: ... ok
test_unauthenticated_user_is_redirected (blood_search.tests.BloodAvailabilitySearchTests.test_unauthenticated_user_is_redirected)
TC-SEC-01 (search part) / SEC-01: ... ok
test_unit_expiring_today_is_still_available (blood_search.tests.BloodAvailabilitySearchTests.test_unit_expiring_today_is_still_available)
Boundary: a unit whose expiry date is today has not expired yet. ... ok
test_user_without_role_is_forbidden (blood_search.tests.BloodAvailabilitySearchTests.test_user_without_role_is_forbidden)
SEC-02: A logged-in user with no BBMS role cannot search. ... ok
test_accepted_donation_details_are_available_for_inventory (donations.tests.DonationHistoryLinkTests.test_accepted_donation_details_are_available_for_inventory) ... ok
test_donation_is_linked_to_correct_donor (donations.tests.DonationHistoryLinkTests.test_donation_is_linked_to_correct_donor) ... ok
test_accepted_donation_immediately_updates_inventory (donations.tests.ImmediateInventoryReflectionTests.test_accepted_donation_immediately_updates_inventory) ... ok
test_inventory_dashboard_refresh_shows_latest_quantity (donations.tests.ImmediateInventoryReflectionTests.test_inventory_dashboard_refresh_shows_latest_quantity) ... ok
test_staff_views_correct_donor_donation_history (donors.tests.DonorHistoryTests.test_staff_views_correct_donor_donation_history) ... ok
test_invalid_donor_registration (donors.tests.DonorRegistrationTests.test_invalid_donor_registration)
TC-DON-02: ... ok
test_mandatory_fields_are_required (donors.tests.DonorRegistrationTests.test_mandatory_fields_are_required)
FR-1.2: ... ok
test_non_blood_bank_staff_cannot_register_donor (donors.tests.DonorRegistrationTests.test_non_blood_bank_staff_cannot_register_donor)
FR-1.1: ... ok
test_unauthenticated_user_cannot_register_donor (donors.tests.DonorRegistrationTests.test_unauthenticated_user_cannot_register_donor)
FR-1.1: ... ok
test_valid_donor_registration (donors.tests.DonorRegistrationTests.test_valid_donor_registration)
TC-DON-01: ... ok
test_non_blood_bank_staff_cannot_search_donors (donors.tests.DonorSearchUpdateTests.test_non_blood_bank_staff_cannot_search_donors)
FR-1.4: ... ok
test_non_blood_bank_staff_cannot_update_donor (donors.tests.DonorSearchUpdateTests.test_non_blood_bank_staff_cannot_update_donor)
FR-1.5: ... ok
test_staff_can_search_donor_by_phone (donors.tests.DonorSearchUpdateTests.test_staff_can_search_donor_by_phone)
TC-DON-03 / FR-1.4: ... ok
test_staff_can_update_donor_address (donors.tests.DonorSearchUpdateTests.test_staff_can_update_donor_address)
TC-DON-03 / FR-1.5: ... ok
test_available_units_exclude_expired_donations (inventory.tests.InventoryServiceTests.test_available_units_exclude_expired_donations) ... ok
test_expired_donations_are_reported (inventory.tests.InventoryServiceTests.test_expired_donations_are_reported) ... ok
test_inventory_contains_all_blood_groups (inventory.tests.InventoryServiceTests.test_inventory_contains_all_blood_groups) ... ok
test_low_stock_groups_use_fewer_than_five_units (inventory.tests.InventoryServiceTests.test_low_stock_groups_use_fewer_than_five_units) ... ok
test_near_expiry_donations_within_seven_days (inventory.tests.InventoryServiceTests.test_near_expiry_donations_within_seven_days) ... ok
test_blood_search_stock_drops_after_issue (issues.tests.BloodSearchReflectsIssuedStockTests.test_blood_search_stock_drops_after_issue) ... ok
test_anonymous_user_gets_401_and_nothing_is_issued (issues.tests.IssueApiTests.test_anonymous_user_gets_401_and_nothing_is_issued) ... ok
test_get_is_not_allowed (issues.tests.IssueApiTests.test_get_is_not_allowed) ... ok
```

```
test_hospital_user_gets_403_and_nothing_is_issued (issues.tests.IssueApiTests.test_hospital_user_gets_403_and_nothing_is_issued) ... ok
test_insufficient_stock_returns_409_and_request_stays_approved (issues.tests.IssueApiTests.test_insufficient_stock_returns_409_and_request_stays_approved) ... ok
test_missing_request_id_returns_400 (issues.tests.IssueApiTests.test_missing_request_id_returns_400) ... ok
test_pending_request_returns_409_and_issues_nothing (issues.tests.IssueApiTests.test_pending_request_returns_409_and_issues_nothing) ... ok
test_route_is_exactly_api_issues (issues.tests.IssueApiTests.test_route_is_exactly_api_issues) ... ok
test_second_post_for_same_request_returns_409 (issues.tests.IssueApiTests.test_second_post_for_same_request_returns_409) ... ok
test_session_post_without_csrf_token_is_rejected (issues.tests.IssueApiTests.test_session_post_without_csrf_token_is_rejected) ... ok
test_staff_can_issue_with_form_data (issues.tests.IssueApiTests.test_staff_can_issue_with_form_data) ... ok
test_staff_can_issue_with_json_body (issues.tests.IssueApiTests.test_staff_can_issue_with_json_body) ... ok
test_unknown_request_returns_404 (issues.tests.IssueApiTests.test_unknown_request_returns_404) ... ok
test_allocation_failure_rolls_back_entire_transaction (issues.tests.IssueBloodServiceTests.test_allocation_failure_rolls_back_entire_transaction) ... ok
test_cannot_issue_the_same_request_twice (issues.tests.IssueBloodServiceTests.test_cannot_issue_the_same_request_twice) ... ok
test_expired_units_are_never_issued (issues.tests.IssueBloodServiceTests.test_expired_units_are_never_issued) ... ok
test_insufficient_stock_changes_nothing (issues.tests.IssueBloodServiceTests.test_insufficient_stock_changes_nothing) ... ok
test_issue_marks_request_fulfilled (issues.tests.IssueBloodServiceTests.test_issue_marks_request_fulfilled) ... ok
test_issue_records_who_issued_it (issues.tests.IssueBloodServiceTests.test_issue_records_who_issued_it) ... ok
test_issue_reduces_available_stock (issues.tests.IssueBloodServiceTests.test_issue_reduces_available_stock) ... ok
test_pending_donations_are_not_issued (issues.tests.IssueBloodServiceTests.test_pending_donations_are_not_issued) ... ok
test_pending_request_cannot_be_issued (issues.tests.IssueBloodServiceTests.test_pending_request_cannot_be_issued) ... ok
test_rejected_request_cannot_be_issued (issues.tests.IssueBloodServiceTests.test_rejected_request_cannot_be_issued) ... ok
test_stock_never_goes_negative_across_issues (issues.tests.IssueBloodServiceTests.test_stock_never_goes_negative_across_issues) ... ok
test_units_come_from_soonest_expiring_batch_first (issues.tests.IssueBloodServiceTests.test_units_come_from_soonest_expiring_batch_first) ... ok
test_anonymous_user_is_redirected_to_login (issues.tests.IssueViewTests.test_anonymous_user_is_redirected_to_login) ... ok
test_hospital_user_cannot_issue_blood (issues.tests.IssueViewTests.test_hospital_user_cannot_issue_blood) ... ok
test_hospital_user_cannot_open_issue_page (issues.tests.IssueViewTests.test_hospital_user_cannot_open_issue_page) ... ok
test_issue_page_shows_batches_after_issue (issues.tests.IssueViewTests.test_issue_page_shows_batches_after_issue) ... ok
test_opening_issue_page_does_not_issue_blood (issues.tests.IssueViewTests.test_opening_issue_page_does_not_issue_blood) ... ok
test_post_for_pending_request_shows_error_and_issues_nothing (issues.tests.IssueViewTests.test_post_for_pending_request_shows_error_and_issues_nothing) ... ok
test_staff_can_open_issue_page (issues.tests.IssueViewTests.test_staff_can_open_issue_page) ... ok
test_staff_post_issues_blood_and_redirects (issues.tests.IssueViewTests.test_staff_post_issues_blood_and_redirects) ... ok
test_allows_issue_when_requested_equals_available (issues.tests.StockValidationTests.test_allows_issue_when_requested_equals_available) ... ok
test_allows_issue_when_stock_is_sufficient (issues.tests.StockValidationTests.test_allows_issue_when_stock_is_sufficient) ... ok
test_rejects_issue_when_no_stock_is_available (issues.tests.StockValidationTests.test_rejects_issue_when_no_stock_is_available) ... ok
test_rejects_issue_when_stock_is_insufficient (issues.tests.StockValidationTests.test_rejects_issue_when_stock_is_insufficient) ... ok
test_rejects_negative_requested_units (issues.tests.StockValidationTests.test_rejects_negative_requested_units) ... ok
test_rejects_zero_requested_units (issues.tests.StockValidationTests.test_rejects_zero_requested_units) ... ok
test_administrator_can_open_reports (reports.tests.ReportsAccessTests.test_administrator_can_open_reports) ... ok
test_anonymous_user_is_redirected_to_login (reports.tests.ReportsAccessTests.test_anonymous_user_is_redirected_to_login) ... ok
test_blood_bank_staff_can_open_reports (reports.tests.ReportsAccessTests.test_blood_bank_staff_can_open_reports) ... ok
test_hospital_user_cannot_open_reports (reports.tests.ReportsAccessTests.test_hospital_user_cannot_open_reports) ... ok
test_group_row_shows_current_stock_after_issue (reports.tests.ReportsContentTests.test_group_row_shows_current_stock_after_issue) ... ok
test_issue_history_lists_the_issue (reports.tests.ReportsContentTests.test_issue_history_lists_the_issue) ... ok
test_patient_names_are_not_shown (reports.tests.ReportsContentTests.test_patient_names_are_not_shown) ... ok
test_request_history_shows_status (reports.tests.ReportsContentTests.test_request_history_shows_status) ... ok
test_summary_counts_donated_and_issued_units (reports.tests.ReportsContentTests.test_summary_counts_donated_and_issued_units) ... ok
test_blood_group_filter (reports.tests.ReportsFilterTests.test_blood_group_filter) ... ok
test_donations_outside_date_range_are_excluded (reports.tests.ReportsFilterTests.test_donations_outside_date_range_are_excluded) ... ok
test_end_date_before_start_date_shows_error (reports.tests.ReportsFilterTests.test_end_date_before_start_date_shows_error) ... ok
test_request_status_filter (reports.tests.ReportsFilterTests.test_request_status_filter) ... ok
test_core_flow_in_each_browser (blood_requests.test_browser_portability.BrowserPortabilityTests.test_core_flow_in_each_browser)
TC-PORT-01 / NFR-08: the core flow works in every listed browser. ... skipped 'Set BROWSER_TESTS=chrome,msedge,firefox and install playwright to run.'
test_concurrent_issues_do_not_overspend_stock (issues.tests.ConcurrentIssuanceTests.test_concurrent_issues_do_not_overspend_stock) ... skipped 'This concurrency test requires PostgreSQL.'

----------------------------------------------------------------------
Ran 131 tests in 74.701s

OK (skipped=2)
Destroying test database for alias 'default' ('file:memorydb_default?mode=memory&cache=shared')...
```